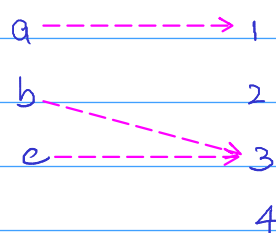


Domain Range



Functional Dependency

```

y = square(x)
int square (int x)
{
    return x * x;
}
    
```

If for every unique value of X
 I get the same value for Y
 the X functionally defines Y
 or Y is functionally dependent on X
 $X \rightarrow Y$

x	y
3	9
4	16
3	9
5	25
-3	9
8	64
1	1
-6	36
-1	1
0	0

attributes

A	B	C	D	E
a1	b1	c1	d1	e1
a2	b1	C2	d2	e1
a3	b2	C1	d1	e1
a4	b2	C2	d2	e1
a5	b3	C3	d1	e1

records

$CD \rightarrow B$ X

$BC \rightarrow D$ ✓

$AC \rightarrow D$ ✓

$AE \rightarrow B$ ✓

Table R

$C \rightarrow D$ ✓

$D \rightarrow C$ X

$D \rightarrow E$ ✓

$B \rightarrow A$ X

$A \rightarrow B$ ✓

Armstrong Axioms

- ① If $B \subseteq A$ then
 $A \rightarrow B$

Reflexivity

$$A \rightarrow A$$

name, age, address $\rightarrow$ age

- ② If $A \rightarrow B$ then

$$AX \rightarrow BX$$

Augmentation

name $\rightarrow$ sex

name, age $\rightarrow$ sex, age

- ③ If $A \rightarrow B$ & $B \rightarrow C$ then

$$A \rightarrow C$$

Transitivity

roll # $\rightarrow$ address

address $\rightarrow$ telephone

roll # $\rightarrow$ telephone

- ④ If $A \rightarrow B$ & $A \rightarrow C$ then

$$A \rightarrow BC$$

Union

roll # $\rightarrow$ name

roll # $\rightarrow$ address

roll # $\rightarrow$ name, address

- ⑤ If $A \rightarrow B$ & $X \rightarrow Y$ then

$$AX \rightarrow BY$$

Composition

roll # $\rightarrow$ s_name

cid $\rightarrow$ t_name

roll #, cid $\rightarrow$ s_name, t_name

- ⑥ If $AX \rightarrow BY$ then

$$AX \rightarrow B \quad \& \quad AX \rightarrow Y$$

Decomposition

reverse of Union

roll #, cid $\rightarrow$ s_name, t_name, grade

roll #, cid $\rightarrow$ s_name

roll #, cid $\rightarrow$ t_name

roll #, cid $\rightarrow$ grade

- ⑦ If $A \rightarrow B$ & $BC \rightarrow D$ then

$$AC \rightarrow D$$

Pseudo Transitivity

$R(A, B, C, D, E, F)$

$A \rightarrow B \quad B \rightarrow C \quad C \rightarrow D \quad E \rightarrow F \quad F \rightarrow E$

$A \rightarrow A$ reflexivity

$A \rightarrow B$ given

$A \rightarrow C$ $A \rightarrow B, B \rightarrow C$ transitivity

$A \rightarrow D$ transitivity $A \rightarrow B, B \rightarrow C, C \rightarrow D$

$A \rightarrow E$ not possible

$AF \rightarrow E$

$A \rightarrow A$ reflexivity

$F \rightarrow E$ given

$AF \rightarrow AE$ composition

$AF \rightarrow A$
 $AF \rightarrow E$ } decomposition

$AE \rightarrow ABCDEF$

$A \rightarrow A$ reflexivity

$A \rightarrow B$ given

$A \rightarrow C$ transitivity

$A \rightarrow D$ "

$A \rightarrow ABCD$ union

$E \rightarrow E$ reflexivity

$E \rightarrow F$ given

$AE \rightarrow ABCDEF$ composition

Closure

$R(A \ B \ C \ D \ E)$

$A \rightarrow BC$

$C \rightarrow B$

$D \rightarrow E$

$E \rightarrow D$

①

$A^+ \rightarrow ABC$

$B^+ \rightarrow B$

$C^+ \rightarrow CB$

$D^+ \rightarrow DE$

$E^+ \rightarrow ED$

②

$AB^+ \rightarrow ABC$

$AC^+ \rightarrow ABC$

$AD^+ \rightarrow ABCDE$

$AE^+ \rightarrow ABCDE$

BC^+

BD^+

BE^+

CD^+

CE^+

DE^+

③

ABC^+

ABD^+

ABE^+

ACD^+

$ACE^+ \rightarrow ADEBC$

ADE^+

BCD^+

BCE^+

BDE^+

CDE^+

④

$ABCD^+$

$ABCE^+$

$ABDE^+$

$ACDE^+$

$BCDE^+$

⑤

$ABCDE \rightarrow ABCDE$

Superkey: Any set of attributes which can uniquely identify each record

of possible FD = $2^n - 1$

5 attributes = $2^5 - 1 = 31$

6 " = $2^6 - 1 = 63$

HW-1

① $R(A B C D E)$
 $A \rightarrow BC \quad C \rightarrow B \quad D \rightarrow E \quad E \rightarrow D$

② $R(A B C D E)$
 $A \rightarrow B \quad B \rightarrow C \quad D \rightarrow C$

Find closure of all possible combinations of attributes using the given functional dependencies

Also identify all super keys

Handwritten submission

at start of class on 17th June 2022

$R(A \ B \ C \ D \ E) \quad A \rightarrow BC \quad C \rightarrow B \quad D \rightarrow E \quad E \rightarrow D$

$SK = \{ \underset{(1)}{AD}, \underset{(2)}{AE}, \underset{(3)}{ABD}, \underset{(3)}{ABE}, \underset{(3)}{ACD}, \underset{(3)}{ACE}, \underset{(3)}{ADE}, \underset{(4)}{ABCD}, \underset{(4)}{ABCE}, \underset{(4)}{ABDE}, \underset{(4)}{ACDE}, \underset{(5)}{ABCDE} \}$

Superkey: Any set of attribute that can identify the whole record.

Default Superkey: Set of all attributes of the Relation

Size of Superkey: # of attributes in the superkey

Candidate key: Superkeys with the smallest size.

$C.K = \{AD, AE\}$

Primary key: One of the candidate keys

$R(A \ B \ C \ D \ E) \quad AB \rightarrow C \quad C \rightarrow E \quad D \rightarrow A$

Find the candidate keys for this relation

$C \rightarrow E$
 $AB \rightarrow C$
 $D \rightarrow A$

$A^+ \rightarrow A$

$AB^+ \rightarrow ABCE$

$BD^+ \rightarrow BDACE \checkmark$

$B^+ \rightarrow B$

$AC^+ \rightarrow ACE$

$BE^+ \rightarrow BE$

$C^+ \rightarrow CE$

$AD^+ \rightarrow AD$

$CD^+ \rightarrow CDAE$

$D^+ \rightarrow DA$

$AE^+ \rightarrow AE$

$CE^+ \rightarrow CE$

$E^+ \rightarrow E$

$BC^+ \rightarrow BCE$

$DE^+ \rightarrow DEA$

$CK = \{BD\}$

$PK = \{BD\}$

$R(W \ X \ Y \ Z)$

$Z \rightarrow WXY$

$X \rightarrow W$

$W^+ \rightarrow W$

$X^+ \rightarrow XW$

$Y^+ \rightarrow Y$

$Z^+ \rightarrow ZWXY \checkmark$

$CK = \{Z\}$

L	B	R
Z	X	W Y

$\overline{D}$	D
Z	W X Y

$R(A \ B \ C \ D \ E \ F)$

$A \rightarrow C$

$C \rightarrow B$

$B \rightarrow D$

$D \rightarrow E$

$A^+ \rightarrow ACBDE \checkmark$

$B^+ \rightarrow BDE$

$C^+ \rightarrow CBDE$

$D^+ \rightarrow DE$

$E^+ \rightarrow E$

$CK = \{A\}$

L	B	R
A	B C D	E

$\overline{D}$	D
A	B C D E

$R(A \ B \ C \ D \ E)$

$A \rightarrow C$

$E \rightarrow D$

$B \rightarrow C$

$ABE^+ \rightarrow ABEC D \checkmark$

$CK = \{ABE\}$

L	B	R
A		C
B		D
E		

$\overline{D}$	D
A	C
B	D
E	

$R(V \ W \ X \ Y \ Z)$

$Z \rightarrow WXY$

$X \rightarrow W$

$Y \rightarrow Z$

$V^+ \rightarrow V$

$W^+ \rightarrow W$

$VW^+ \rightarrow VW$

$VZ^+ \rightarrow VZWXY \checkmark$

$VX^+ \rightarrow VXW$

$VX^+ \rightarrow VXW$

$VY^+ \rightarrow VYZWX \checkmark$

$VY^+ \rightarrow VYZWX \checkmark$

$VZ^+ \rightarrow VZWXY \checkmark$

$CK = \{VY, VZ\}$

P.A. = $\{V, Y, Z\}$

NPA = $\{W, X\}$

$\overline{D}$	D
V	W X Y Z

L	B	R
V	Z X Y	W

$R(A\ B\ C\ D\ E)$ $A \rightarrow C$ $C \rightarrow B$ $B \rightarrow D$ $D \rightarrow E$ $E \rightarrow A$

$(\{A\} \cup \{A\})^+ = A^+ \rightarrow ABCDE \checkmark$
 $B^+ \rightarrow ABCDE \checkmark$
 $C^+ \rightarrow ABCDE \checkmark$
 $D^+ \rightarrow ABCDE \checkmark$
 $E^+ \rightarrow ABCDE \checkmark$

L	B	R
	A	
	B	
	C	
	D	
	E	

<u>D</u>	D
	A
	B
	C
	D
	E

CK: $\{A, B, C, D, E\}$

P.A: $\{A, B, C, D, E\}$

N.P.A: $\{\}$

$R(A\ B\ C\ D\ E\ F\ G\ H\ I\ J)$

$AB \rightarrow EF$ $A \rightarrow C$ $B \rightarrow DGH$ $H \rightarrow I$ $AE \rightarrow BDFGHI$

$AJ^+ \rightarrow ASC$

$AJB^+ \rightarrow ASBCEFDGHI \checkmark$

$AJE^+ \rightarrow AJECBDFGHI \checkmark$

$AJH^+ \rightarrow AJHCI$

L	B	R
A	B	C
J	E	D
	H	F
		G
		I

CK: $\{AJB, AJE\}$

P.A: $\{A, J, B, E\}$

NPA: $\{C, D, F, G, H, I\}$

Prime Attributes: Any attribute that is part of any candidate key

Non-Prime Attributes: Any attribute that is not a prime attribute

12. $R(A B C D E F G H)$

$AB \rightarrow C \quad AC \rightarrow B \quad AD \rightarrow E \quad B \rightarrow D \quad BC \rightarrow A \quad E \rightarrow G$

$FH^+ \rightarrow FH$

$FHA^+ \rightarrow FHA$

$FHB^+ \rightarrow FHB$

$FHC^+ \rightarrow FHC$

$FHD^+ \rightarrow FHD$

$FHE^+ \rightarrow FHE$

$FHAB \rightarrow FHAB C D E G \checkmark$

$FHAC \rightarrow FHAC B D E G \checkmark$

$FHAD \rightarrow FHADEG$

$FHAE$

$FHBC^+ \rightarrow FHBC D A E G \checkmark$

$FHBD$

$FHBE$

$FHCD$

$FHCE$

$FHDE$

L	B	R
F	A	G
H	B	
	C	
	D	
	E	

$CK = \{FHAB, FHAC, FHBC \dots\}$

$P.A = \{A, B, C, F, H\}$

$NPA = \{D, E, G\}$

Normalization

0-Normal Form

You must have a table with data

1st Normal Form

System must be in 0-NF

AND

No Multivalued or Complex Attributes allowed

OR

System should have FDs

OR

System should have a PK.

2nd - Normal Form

System must be in 1st NF

AND

All non prime attributes must be functionally dependant on the whole primary key

OR

No partial dependency between Non Prime Attributes and the PK is allowed

3rd Normal Form

System must be in 2nd NF

AND

All non prime attributes must be directly functionally dependant on the PK

OR

No transitive dependency is allowed

only applies
to relations
with
Composite PK

Item	Color	Price	Tax	Qty
T-Shirt	Red, Blue	12.00	0.60	5, 3
Polo	Red, Yellow	15.00	0.60	8, 2
T-Shirt	Red, Black	12.00	0.60	5, 9
Sweatshirt	Blue, Black	25.00	1.25	12, 2

<u>Item</u>	<u>Color</u>	Price	Tax	Qty
T-shirt	Red	12.00	0.60	5
T-shirt	Blue	12.00	0.60	3
Polo	Red	15.00	0.60	8
Polo	Yellow	15.00	0.60	2
T-shirt	Red	12.00	0.60	5
T-shirt	Black	12.00	0.60	9
S-shirt	Blue	25.00	1.25	12
S-shirt	Black	25.00	1.25	2

price $\rightarrow$ tax

item $\rightarrow$ price

item, color $\rightarrow$ qty

L	B	R
item	price	tax
color		qty

$(\text{item, color})^+ \rightarrow \text{item, color, price, tax, qty}$

$CK = \{(\text{item, color})\}$

$PK = (\text{item, color})$

$R1(\underline{\text{item}}, \underline{\text{color}}, \text{price}, \text{tax}, \text{qty})$

$R1$ is in 1-NF

PK = ABC

A AB
B AC
C BC

P.A = {item, color}

N.P.A = {~~price~~, ~~tax~~, qty}

item → price
price → tax

L	B	R
I	P	T

item⁺ → item, price, tax

color⁺ → color

R2 (item, price, tax)

R1 (item, color, ~~price~~, ~~tax~~, qty)

R4 (item, color, qty)

R2 & R4 are in 2nd NF

R2

item	price	tax
T-shirt	12.00	0.60
Polo	15.00	0.60
S-shirt	25.00	1.25

R3

item	color	qty
T-shirt	Red	5
T-shirt	Blue	3
Polo	Yellow	2
Polo	Red	8
T-shirt	Black	9
S-shirt	Blue	12
S-shirt	Black	2

R2 (item, price, tax)

R3 (item, color, qty)

R4 (item, price)

R5 (price, tax)

R3, R4, R5 are in 3NF

R3

item	color	qty
T-shirt	Red	5
T-shirt	Blue	3
Polo	Yellow	2
Polo	Red	8
T-shirt	Black	9
S-shirt	Blue	12
S-shirt	Black	2

R4

item	price
T-shirt	12.00
Polo	15.00
S-shirt	25.00

R5

price	tax
12.00	0.60
15.00	0.60
25.00	1.25

$R(A\ B\ C\ D\ E\ F\ G\ H\ I\ J\ K\ L\ M)$

$A \rightarrow CE\quad B \rightarrow DF\quad AB \rightarrow G\quad G \rightarrow BI\quad AG \rightarrow H\quad AJ \rightarrow K\quad C \rightarrow L$
 $AJB \rightarrow M$

1st NF

$AJ^+ \rightarrow AJCELK$

$AJB^+ \rightarrow AJCELKBDFGIHM \checkmark$

$AJC^+ \rightarrow AJCELK$

$AJG^+ \rightarrow AJCELKG B I D F M H \checkmark$

L	B	R
A	B	D
J	C	E
	G	F
		H
		I
		K
		L
		M

$CK = \{ \textcircled{AJB}\ AJG \}$

$R1(\underline{A}\ \underline{B}\ C\ D\ E\ F\ G\ H\ I\ \underline{J}\ K\ L\ M)$

$R1$ is in 1-NF

2nd Normal Form

$P.A = \{A, J, B, G\}$

$N.P.A = \{ \cancel{C}, \cancel{D}, \cancel{E}, \cancel{F}, \cancel{H}, \cancel{I}, \cancel{K}, \cancel{L}, \cancel{M} \}$

$Pk = AJB$

$A^+ \rightarrow ACEL$

$R2(\underline{A}CEL)$

$J^+ \rightarrow J$

$B^+ \rightarrow BDF$

$R3(\underline{B}DF)$

$AJ^+ \rightarrow AJCELK$

$R4(\underline{A}\ \underline{J}\ K)$

$AB^+ \rightarrow AB \cancel{C} \cancel{E} \cancel{L} \cancel{D} \cancel{F} G I H$

$R5(\underline{A}\ \underline{B}\ G\ I\ H)$

$JB^+ \rightarrow JB \cancel{D} \cancel{F}$

$R6(\underline{A}\ \underline{J}\ \underline{B}\ M)$

$R2, R3, R4, R5, R6$ are in 2nd NF

$W \rightarrow x \quad x \rightarrow y \quad y \rightarrow z$
 $R(\underline{W}, \underline{x}, \underline{y}, \underline{z}) \quad T_1(\underline{W}, x) \quad T_2(x, y) \quad T_3(y, z)$

3rd NF

$R_2(\underline{A} \underline{C} E L)$

$R_7(\underline{A} C E) \quad R_8(\underline{C} L)$

$R_3(\underline{B} \underline{D} F)$

$R_4(\underline{A} \underline{J} K)$

$AB \rightarrow G$
 $AG \rightarrow H$
 $\{A\} \cup \{AB\} \xrightarrow{G} H$
 $\downarrow$
 $AB \rightarrow H$

$R_5(\underline{A} \underline{B} G I H)$

$AB \rightarrow G$
 $G \rightarrow B$
 $G \rightarrow I$
 $AG \rightarrow H$

$R_9(\underline{A} \underline{B} G H) \quad R_{10}(G I)$

$R_6(\underline{A} \underline{J} \underline{B} M)$

$R_3, R_4, R_6, R_7, R_8, R_9, R_{10}$ are in 3NF

Boyce Codd Normal Form (BCNF)

If you have an FD for the relation R of the form

$\alpha \rightarrow \beta$

then either

(i) this is a trivial FD

or (ii) α is a Superkey of R

	non trivial	is LHS a SK
$A \rightarrow C$	✓	✓ (R7)
$A \rightarrow E$	✓	✓ (R7)
$B \rightarrow D$	✓	✓ (R3)
$B \rightarrow F$	✓	✓ (R3)
$AB \rightarrow G$	✓	✓ (R9)
$AG \rightarrow H$	✓	✓ (R9)
$G \rightarrow B$	✓	X
$G \rightarrow I$	✓	✓ (R10)
$AJ \rightarrow K$	✓	✓ (R4)
$C \rightarrow L$	✓	✓ (R8)
$AJB \rightarrow M$	✓	✓ (R6)

$R_3(\underline{B} D F)$

$R_4(\underline{A} \underline{J} K)$

$R_6(\underline{A} \underline{J} \underline{B} M)$

$R_7(\underline{A} C E)$

$R_8(\underline{C} L)$

~~$R_9(\underline{A} \underline{B} G H)$~~

$R_{10}(\underline{G} I)$

~~$AG^+ \rightarrow AG \{C, L, B, D, F, H\}$~~

$AG^+ \rightarrow AG B H$

$R_{11}(\underline{G} B)$

$R_{12}(\underline{A} \underline{G} H)$

$R(C^{\overset{A}{\#}}, C^{\overset{B}{\text{Name}}}, C^{\overset{C}{\text{Address}}}, C^{\overset{D}{\text{Phone}}}, E^{\overset{E}{\#}}, E^{\overset{F}{\text{Name}}}, E^{\overset{G}{\text{Address}}}, E^{\overset{H}{\text{Phone}}}, E^{\overset{I}{\text{Salary}}}, P^{\overset{J}{\text{No}}}, P^{\overset{K}{\text{Name}}}, P^{\overset{L}{\text{Desc}}}, P^{\overset{M}{\text{Price}}}, P^{\overset{N}{\text{O\#}}}, P^{\overset{O}{\text{Date}}}, P^{\overset{P}{\text{Qty}}})$

$C^{\overset{A}{\#}} \rightarrow C^{\overset{B}{\text{Name}}}, C^{\overset{C}{\text{Address}}}, C^{\overset{D}{\text{Phone}}}$

$E^{\overset{E}{\#}} \rightarrow E^{\overset{F}{\text{Name}}}, E^{\overset{G}{\text{Address}}}, E^{\overset{H}{\text{Phone}}}, E^{\overset{I}{\text{Salary}}}$

$P^{\overset{J}{\text{No}}} \rightarrow P^{\overset{K}{\text{Name}}}, P^{\overset{L}{\text{Desc}}}, P^{\overset{M}{\text{Price}}}$

$P^{\overset{N}{\text{O\#}}} \rightarrow P^{\overset{O}{\text{Date}}}, C^{\overset{A}{\#}}, E^{\overset{E}{\#}}$

$P^{\overset{N}{\text{O\#}}}, P^{\overset{J}{\text{No}}} \rightarrow P^{\overset{P}{\text{Qty}}}$

$A \rightarrow BCD$

$E \rightarrow FGHI$

$J \rightarrow KLM$

$N \rightarrow OAE$

$NI \rightarrow P$

$(JN)^+ \rightarrow JNKLMOAEBGHI P \quad \checkmark$

L	B	R
J	A	B F K
N	E	C G L
		D H M
		I O P

$CK = \{(JN)^+\}$

$R(A B C D E F G H I \underline{J} K L M \underline{N} O P) \quad \underline{1NF}$

$P.A = \{J, N\}$

$NPA = \{\cancel{A} \cancel{B} \cancel{C} \cancel{D} \cancel{E} \cancel{F} \cancel{G} \cancel{H} \cancel{I} \cancel{K} \cancel{L} \cancel{M} \cancel{O} \cancel{P}\}$

$J^+ \rightarrow JKLM$

$R_2(\underline{J} K L M)$

$N^+ \rightarrow NOAEBGHI$

$R_3(A B C D E F G H I \underline{N} O)$

$R_4(\underline{J} \underline{N} P)$

$\left. \begin{array}{l} R_2(\underline{J} K L M) \\ R_3(A B C D E F G H I \underline{N} O) \\ R_4(\underline{J} \underline{N} P) \end{array} \right\} 2NF$

$R_2(\underline{J} K L M)$

$R_3(A B C D E F G H I \underline{N} O)$

$R_4(\underline{J} \underline{N} P)$

$R_2(\underline{J} K L M)$

$R_4(\underline{J} \underline{N} P)$

$R_5(\underline{A} B C D)$

$R_6(\underline{E} F G H I)$

$R_7(\underline{N} O A E)$

$\left. \begin{array}{l} R_5(\underline{A} B C D) \\ R_6(\underline{E} F G H I) \\ R_7(\underline{N} O A E) \end{array} \right\} 3NF$

is Superkey

$A \rightarrow B$

✓

$A \rightarrow C$

✓

$A \rightarrow D$

✓

$E \rightarrow F$

✓

$E \rightarrow G$

✓

$E \rightarrow H$

✓

$E \rightarrow I$

✓

$J \rightarrow K$

✓

$J \rightarrow L$

✓

$J \rightarrow M$

✓

$N \rightarrow O$

✓

$N \rightarrow A$

✓

$N \rightarrow E$

✓

$JN \rightarrow P$

✓

BCNF

$R2(PNo, PName, PDesc, Pprice)$

$R4(PNo, O\#, Qty)$

$R5(C\#, CName, CAddress, CPhone)$

$R6(E\#, EName, EAddress, EPhone, ESalesy)$

$R7(O\#, ODate, C\#, E\#)$

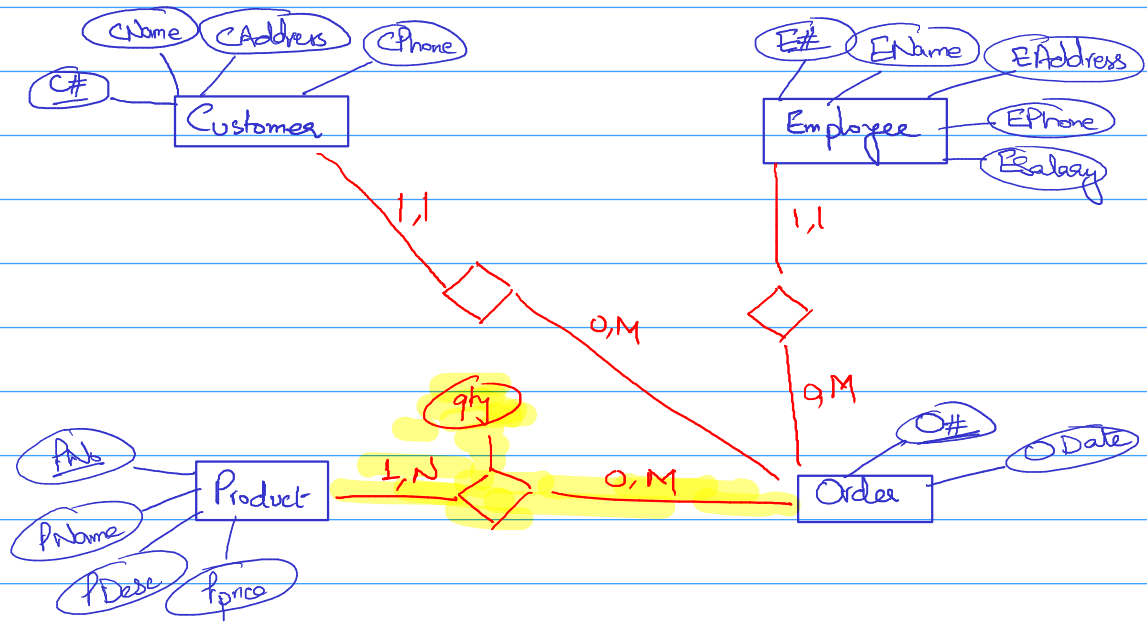
$R2(\underline{J}KLM)$

$R4(\underline{J}NP)$

$R5(\underline{A}BCD)$

$R6(\underline{E}FGHI)$

$R7(\underline{N}OAE)$



$$\alpha \rightarrow \beta$$

$$A \rightarrow B$$

$$A \rightarrow CD$$

$$AX \rightarrow E$$

$$AX \rightarrow FG$$

$$Y \rightarrow Z$$

$$Z \rightarrow B$$

$$\alpha : \{SK, CK, P, PC, NP, NPC\}$$

$$\beta : \{P, NP\}$$

$$A \rightarrow B$$

$$A \rightarrow C$$

$$A \rightarrow D$$

$$AX \rightarrow E$$

$$AX \rightarrow F$$

$$AX \rightarrow G$$

$$XY \rightarrow A$$

$$F \rightarrow A$$

$$AB \rightarrow CDEF$$

let $\{A, B\}$ be

prime attributes

& let F be a

non. prime attribute

$$(AB \rightarrow CDEF)$$

$$(F \rightarrow A)B \rightarrow CDEF$$

then

FB will be a SK

thus FB will also be

CK

& F must be prime

XYB is a SK

	BENF	3NF	2NF	1NF
$SK \rightarrow P$	✓	✓	✓	✓
$SK \rightarrow NP$	✓	✓	✓	✓
$CK \rightarrow P$	✓	✓	✓	✓
$CK \rightarrow NP$	✓	✓	✓	✓
$P \rightarrow P$		✓	✓	✓
$P \rightarrow NP$				✓
$PC \rightarrow P$		✓	✓	✓
$PC \rightarrow NP$				✓
$NP \rightarrow P$				
$NP \rightarrow NP$			✓	✓
$NPC \rightarrow P$		✓	✓	✓
$NPC \rightarrow NP$			✓	✓

α must be

SK/CK

α must be

SK/CK

or

β must be prime

Relation has FD

α must be

CK/SK

or

α, β must be prime

or

α, β must be non-prime

$R(A B C D E)$

$AB \rightarrow CD$

$D \rightarrow A$

$BC \rightarrow DE$

$CK = \{AB, BC, BD\}$

$P.A = \{A, B, C, D\}$

$NPA = \{E\}$

L	B	R
B	A	E
	C	
	D	

	BCNF	3NF
$AB \rightarrow C$	✓	✓
$AB \rightarrow D$	✓	✓
$D \rightarrow A$	X	✓
$BC \rightarrow D$		✓
$BC \rightarrow E$		✓

$B^+ \rightarrow B$

$AB^+ \rightarrow ABCDE$ ✓

$BC^+ \rightarrow BCDEA$ ✓

$BD^+ \rightarrow BDACE$ ✓

3NF

$R(A B C D E F G H)$

$AB \rightarrow CD$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$CK = \{AB\}$

$P.A = \{A, B\}$

$N.P.A = \{C, D, E, F, G, H\}$

	BCNF	3NF	2NF	1NF
$AB \rightarrow C$	✓	✓	✓	✓
$AB \rightarrow D$	✓	✓	✓	✓
$A \rightarrow D$	X	X	X	✓
$A \rightarrow E$				✓
$B \rightarrow F$				✓
$F \rightarrow G$				✓
$F \rightarrow H$				✓

1NF